

Cloud Infrastructure & System Administration

Intern

INTERNSHIP VIVA
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Utkarsh Srivastava

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ABSTRACT

This internship at **Ansys Software Pvt Ltd** provided valuable hands-on experience in the field of **Cloud Infrastructure Support using AWS And Azure**. The internship was aimed at enhancing my practical understanding of cloud computing concepts, system administration practices, and enterprise IT infrastructure management.

During the internship, I worked on provisioning and configuring **virtual machines**, managing **cloud storage solutions**, designing **virtual networks**, and setting up **identity and access management policies**. I was also involved in **monitoring system performance**, ensuring **data security and compliance**, and supporting **backup and disaster recovery operations**.

Through active collaboration with the Cloud Infrastructure team, I gained exposure to hybrid cloud environments, automation tools, and best practices in maintaining high availability and scalability. This internship not only strengthened my knowledge of AWS but also improved my problem-solving and technical support skills, which are essential for a career in **Cloud Infrastructure and System Administration**.

INTRODUCTION

Ansys Software Pvt Ltd is a global leader in engineering simulation software, delivering innovative solutions to help customers design . and optimize products. The Cloud Infrastructure & System Administration team plays a critical role in enabling secure and scalable cloud services to support global operations. My role as a System Administrator Intern included contributing to the setup, monitoring, and support of AWS & Azure resources. This internship provided me with exposure to real-world enterprise cloud operations, including hybrid deployments and security management.

Analysis & Requirement

The Cloud Infrastructure Support internship required both technical and analytical skills for managing IT infrastructure in the cloud.

Skills & Knowledge Required:

- Networking fundamentals, Linux/Windows administration, and virtualization.
- AWS services such as EC2, S3, IAM, and VPC.
- Azure services such as Virtual Machines, Blob Storage, Virtual Networks, and Azure Active Directory.

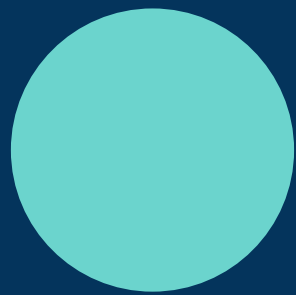
Tools & Technologies Used:

- AWS Console, Azure Portal, CLI, and PowerShell.
- Cloud monitoring tools like AWS CloudWatch and Azure Monitor.

Problem Description

Internship divided into 6 core modules:

- ❖ Identity & Access Management
- ❖ Compute Management
- ❖ Storage & Backup
- ❖ Networking
- ❖ Monitoring & Logging
- ❖ Security & Compliance



Design Phase

Cloud Infrastructure Layout

Organized Identity, Compute, Storage, Networking

Security Architecture

IAM roles, Azure AD groups, MFA policies

Data Flow Design

S3/Blob/EC2/VM communication

Backup & redundancy paths

Monitoring Design

CloudWatch dashboards

Azure Monitor alerts

AWS Implementation

- ✓ Created IAM users, groups, roles
- ✓ Provisioned EC2 instances
- ✓ Configured S3 buckets with lifecycle policies
- ✓ Setup CloudWatch metrics & alarms
- ✓ Automated tasks using AWS CLI
- ✓ Implemented Lambda-based scheduled backups

Azure Implementation

- Created Azure AD users & groups
- Provisioned Virtual Machines
- Configured Blob storage containers
- Setup NSGs & firewall rules
- Configured Azure Monitor dashboards
- Automated tasks with PowerShell scripts

Automation

- Automated server provisioning
- Backup scheduling using **Lambda & Azure Automation**
- CLI & PowerShell for bulk deployments
- Monitoring alerts configured using automated rules

Testing Phase

Functional Testing:

IAM permissions

VM/EC2 connectivity

Backup & restore workflows

Security Testing:

Encryption checks

Pen-testing for open ports

Load Testing:

AWS CloudWatch

Azure Load Testing

CONCLUSION

Internship provided strong foundation in **cloud & system administration**

Gained hands-on experience in **security, monitoring, automation**

Improved troubleshooting, cloud provisioning, and team collaboration skills

Ready for roles in **Cloud Infra, DevOps and System Administration**

Appendices

- Tools & Technologies Used
- AWS Tasks Performed
- Architecture Diagrams
- Scripts & Configurations
- Screenshots

Screenshots

- AWS Console
- EC2 Instances
- S3 Bucket
- Azure Portal
- VM Details

aws

Search

[Alt+S]

Asia Pacific (Sydney)

Account ID: 0603-5170-8303

cloudusrivast

Console home

myApplications

All services

Console home

Info

Reset to default layout

Add widgets

Recently visited

Info

No recently visited services

Explore one of these commonly visited AWS services.

EC2S3Aurora and RDSLambda

View all services

Applications (0)

Info

Create application

Region: Asia Pacific (Sydney)

Select Region

ap-southeast-2 (Current Region)

Find applications

Name

Description

Region

Origin

No applications

Get started by creating an application.

Create application

Go to myApplications

Welcome to AWS

AWS Health

Info

Cost and usage

Info

EC2 Instance

SCREENSHOT

The screenshot shows the AWS Management Console interface for connecting to an EC2 instance. The browser address bar displays the URL: `https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ConnectToInstance:instanceId=i-00c2f19ebc9e503a7`. The console header includes the AWS logo, a search bar, and navigation links for Services, N. Virginia, and the user profile (utksrivast).

The breadcrumb navigation path is: [EC2](#) > [Instances](#) > [i-00c2f19ebc9e503a7](#) > [Connect to instance](#).

Connect to instance [Info](#)

Connect to your instance i-00c2f19ebc9e503a7 (Web Server) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID
i-00c2f19ebc9e503a7 (Web Server)

Connection Type

☒ **Connect using EC2 Instance Connect**
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

☐ **Connect using EC2 Instance Connect Endpoint**
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IP address
54.89.9.24

User name
Enter the user name defined in the AMI used to launch the instance. If you didn't define a custom user name, use the default user name, ubuntu.

Note: In most cases, the default user name, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

Cancel

Connect

Amazon S3 > Buckets > srmusrivast > Upload

Upload

Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 979.0 KB)

All files and folders in this table will be uploaded.

Find by name

☐	Name	▼	Folder	▼	Type	▼	Size
☐	INTERNSHIP - PPT.pptx		-		application/vnd.openxmlformats-o...		979.0 KB

Destination

Info

Destination

[s3://srmusrivast](#)

References

- AWS Documentation
- Azure Documentation
- AWS Well-Architected Framework
- Microsoft Architecture Center
- Ansys Internal Documentation
- Cloud Security Alliance

THANK YOU